

BCA
Third Semester Syllabus 2025-26
University of Mysore

Curriculum

Program: BCA

Subject: Computer Applications

Semester	Course No.	Theory/ Practical	Credits	L-T-P	No. of Hours	Paper Title	Marks	
							SEE	CIE
III	CAM31T	Theory	03	3-0-0	03	C#.NET Programming	80	20
	CAM31P	Practical	02	0-0-2	04	C#.NET Programming Lab	40	10
	CAM32T	Theory	03	3-0-0	03	Data Base Management System	80	20
	CAM32P	Practical	02	0-0-2	04	DBMS Lab	40	10
	CAM33T	Theory	03	3-0-0	03	Web Technologies	80	20
	CAM33P	Practical	02	0-0-2	04	Web Technologies Lab	40	10
	CAE31X	Theory (Elective 1)	03	3-0-0	03	1. Cyber Security 2. Software Engineering 3. System Software	80	20
	CAE32X	Theory (Elective 2)	03	3-0-0	03	1. Cloud Computing 2. Digital Image Processing 3. E-Commerce and E-Governance	80	20

Semester: III

Course Code: CAM31T	Course Title: C# and .NET Programming
Course Credits: 03 (3-0-0)	Hours/Week: 03
Total Contact Hours: 44	Formative Assessment Marks: 20
Exam Marks: 80	Exam Duration: 03

Course Outcomes (COs):

CO1 Understand C# basics, syntax, control statements, arrays, strings.

CO2 Apply OOP concepts like classes, inheritance, and interfaces.

CO3 Use delegates, events, and file I/O in C#.

CO4 Build GUI apps using Windows Forms and ADO.NET.

Course Contents	Hours
Unit 1	
Introduction: Overview of OOP, Introduction to C # - Characteristics, application, origins of the .NET technology, the .NET framework, C# program structure, command line argument, math function, Literals, variable in C#: Declaration, initialization, constant variables, scope of variables, boxing and unboxing, Operators, expression in C#, Decision making and looping statements in C#, Methods in C# : declaring methods, methods parameters:output,ref,val,params, Arrays :Declaration, initialization, variable size arrays, array class, array list class, String handling: introduction, string library functions.	11
Unit-2	
OOPS with C#: Introduction to Classes & Objects, Constructor: Introduction & Types, Destructor, Inheritance & types: single level inheritance ,multilevel inheritance, hierarchical inheritance, containment inheritance, defining a subclass, visibility control, polymorphism: definition, method overloading, method overriding, operator overloading: definition, overloadable operators, overloading unary and binary operator, Data Abstraction: hiding methods, abstract classes, abstract methods, sealed classes, sealed methods, Interfaces: Multiple Inheritance: defining an interface, extending an interface, implementing interface, abstract class and interface.	11

Unit-3	
Delegates and Events: Definition, delegate declaration, delegate methods, delegates instantiation, delegate invocation, multicast delegates, events. Managing Console I/O operations: console class, console input, console output, formatted output, numeric formatting, standard numeric format, custom numeric format. File handling: I/O Classes: write files, read files, File streams: file stream classes, File modes, operations on files	11
Unit-4	
Introducing windows forms: A tale of three GUI namespaces, Anatomy of a Form, Component class, control class. Programming with windows forms controls: Working with button types, check boxes, labels Radio buttons, track bar, Progress bar, Group boxes, list boxes, calendar control, assigning tooltips for controls. Data access with ADO.NET: Introduction, two faces of ADO.NET, role of ADO.NET data providers, building a simple test database, selecting a data provider, working with connected layer of ADO.NET & OleDb Data reader, inserting, updating and deleting records using OleDb command.	11

Text Books:

- 1.Programming in C#, E. Balagurusamy,4th or 5th Edition, McGraw Hill Education.
2. C# 9.0 and .NET 5 – Modern Cross-Platform Development, Mark J. Price,6th Edition, Packt Publishing.

Reference:

- 1.Pro C# 8 with .NET Core 3, Andrew Troelsen and Philip Japikse, Apress Publications.
- 2.Head First C#, Andrew Stellman and Jennifer Greene, O'Reilly Media.
- 3.The Complete Reference C#, Herbert Schildt, McGraw Hill Education.

Course Code: CAM31P	Course Title: C#.NET Programming Lab
Course Credits: 02 (0-0-2)	Hours/Week: 04
Total Contact Hours: 60	Formative Assessment Marks: 10
Exam Marks: 40	Exam Duration: 03

Course Outcomes (COs):

CO1 Demonstrate C# basics using methods, arrays, and strings.

CO2 Apply OOP concepts like inheritance and overloading in C#.

CO3 Implement delegates, events, and file handling in C#.

CO4 Design Windows Forms and connect databases using ADO.NET.

Laboratory Program List

PART -A

1. Write a C# program to add two numbers using command-line arguments.
2. Write a C# program to demonstrate the use of methods and operators.
3. Write a C# program to demonstrate operations on an ArrayList.
4. Write a C# program to demonstrate string functions.
5. Write a C# program to demonstrate default and parameterized constructors using a student class.
6. Write a C# program to demonstrate multilevel inheritance using classes Person, Employee and Manager.
7. Write a C# program to demonstrate method overloading.
8. Write a C# program to overload the + operator to add two objects of a Complex class.

PART-B

1. Write a C# program to implement multicast delegate.
2. Write a C# program to implement an event handler.
3. Write a C# program to demonstrate the operations of numeric formatting types.
4. Write a C# program to write to and read from a text file.
5. Design a Windows Forms application that includes the following controls: a list box, checkboxes, radio buttons, and image button.
6. Create a Windows Forms application that demonstrates the use of a track bar, progress bar, group boxes and button.
7. Create a Windows Forms application that includes a calendar control to allow the user to select a date and display it on a label.
8. Create a Windows Forms application that connects to a test database using ADO.NET with the OleDb provider. Allow the user to insert new records and display existing data using a DataGridView.

Evaluation Scheme for Lab Examination

Assessment Criteria Marks		Marks
Writing	One Program from Part A	15
	One Program from Part B	15
Execution	Any one of the Written Program	5
Viva Voce based on C#.NET Programming		5
Total		40

Semester: III

Course Code: CAM32T	Course Title: Data Base Management System
Course Credits: 03 (3-0-0)	Hours/Week: 03
Total Contact Hours: 44	Formative Assessment Marks: 20
Exam Marks: 80	Exam Duration: 03

Course Outcomes (COs):

CO1: Understand basic concepts of databases, data models, and ER diagrams.

CO2: Apply relational algebra and SQL to query and manipulate data.

CO3: Analyze database normalization to improve design and remove anomalies.

CO4: Evaluate transaction management and database security features.

Content	Hours
Unit-1	
Introduction to Databases: Definition of Data, Database, and DBMS, Overview of Database, Applications, Advantages and Disadvantages of DBMS, Roles of Database Users and Administrators. Data Models: Introduction to Data Models, Types of Data Models (Hierarchical, Network, Relational, Object-oriented), Importance of Data Models in DBMS. Database Design: Keys: Primary Key, Candidate Key, Super Key, Foreign Key, Composite Key, Alternate Key, Unique Key, Surrogate Key, Constraints in a table: Primary Key, Foreign Key, Unique Key, NOT NULL, CHECK, Entity-Relationship (ER) Model, Entities and Entity Sets, Attributes and Relationships, ER Diagrams, Key Constraints and Weak Entity Sets, Extended ER Features, Introduction to the Relational Model and Relational Schema.	11
Unit-2	
Relational Algebra: Introduction to Relational Algebra, Operations: Selection, Projection, Set Operations, Join Operations, Division. Structured Query Language (SQL): SQL Basics: DDL and DML, Aggregate Functions (Min(), Max(), Sum(), Avg(), Count ()), Logical operators (AND, OR, NOT), Predicates (Like, Between, Alias, Distinct), Clauses (Group By, Having, Order by, top/limit).	11

Unit-3	
SQL Joins and Views: Inner Join, Natural Join, Full Outer Join, Left Outer Join, right outer Join, Equi Join, Definition of View, creating a View, Managing Views (Listing, Updating, Deleting). Normalization: Anomalies in relational database design. Functional dependencies - Axioms. Decomposition, Transitive Dependency. Data Normalization: First normal form, Second normal form, Third normal form. Boyce-Codd normal form.	11
Unit-4	
Query Processing Transaction Management: Introduction Transaction Processing, Single user & multiuser systems. Transactions: read & write operations. Need of concurrency control: The lost update problem, Dirty read problem. Types of failures. Transaction states. Desirable properties (ACID properties) of Transactions. Storage of Database, File Operations Database Security.	11

Reference Books:

1. Fundamentals of Database Systems, Ramez Elamassri, Shankant B. Navathe, 7th Edition, Pearson.

Reference Books:

1. An Introduction to Database Systems, Bipin Desai, Galgotia Publications, 2010.
2. Introduction to Database System, C J Date, Pearson, 1999.
3. Database Systems Concepts, Abraham Silberschatz, Henry Korth, S.Sudarshan, 6th Edition, McGraw Hill,
4. Database Management Systems, Raghu Rama Krishnan and Johannes Gehrke, 3rd Edition, McGraw Hill.

Course Code: CAM32P	Course Title: DBMS Lab
Course Credits: 02 (0-0-2)	Hours/Week: 04
Total Contact Hours: 60	Formative Assessment Marks: 10
Exam Marks: 40	Exam Duration: 03

Course Outcomes (COs):

CO1: Execute single-line SQL queries and apply group functions effectively.

CO2: Perform database operations using DDL, DML, DCL, and TCL commands.

CO3: Implement advanced SQL concepts like nested queries and join operations.

CO4: Create views and apply table-level locking mechanisms for data control.

Laboratory Program List

Part A:

Activity 1: Database: Student (DDL, DML Statements)

Table: Student

Name	Reg. No	Class	Major
Smith	17	1	CS
Brown	8	2	CS

Table: Course

Course Name	Course Number	Credit Hours	Department
Introduction to Computer Science	CS1310	4	CS
Data Structure	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database Management System	CS3380	3	CS

Table: Section

Section Identifier	Course Number	Year	Instructor
85	MATH2410	98	King
92	CS1310	98	Andreson
102	CS3320	99	Knuth

112	MATH2410	99	Chang
119	CS1310	99	Andreson
135	CS3380	99	Stone

Table: Grade_Report

Reg. No	Section Identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

Queries

1. Create Table Using create statement.
2. Insert rows into individual Tables using insert statement.
3. Alter table section add new field section and update the records
4. Delete brown's grade report.
5. Drop the table section.

Activity 2: (Select clause, Arithmetic Operators)

Database: Employee

Create the following tables and insert tuples with suitable constraints.

Table: EMPLOYEE

EMPID	FIRSTNAME	LASTNAME	Hire Date	ADDRESS	CITY
1001	George	Smith	11-May-06	83 First Street	Paris
1002	Mary	Jones	25-Feb-08	842 Vine Ave	Losantiville
1012	Sam	Tones	12-Sep-05	33 Elm St.	Paris
1015	Peter	Thompson	19-Dec-06	11 Red Road	Paris
1016	Sarath	Sharma	22-Aug-07	440 MG Road	New Delhi
1020	Monika	Gupta	07-Jun-08	9 Bandra	Mumbai

Table: EMPSALARY

EMPID	SALARY	BENEFITS	DESIGNATION
1001	10000	3000	Manager
002	8000	1200	Salesman
1012	20000	5000	Director
1015	6500	1300	Clerk
1016	6000	1000	Clerk
1020	8000	1200	Salesman

Queries

1. Display FIRSTNAME, LASTNAME, ADDRESS, and CITY of all employees living in PARIS
2. Display the content of the employee table in descending order of FIRSTNAME
3. Select FIRSTNAME and SALARY of salesmen
4. Display the FIRSTNAME, LASTNAME, and TOTAL SALARY of all employees where
TOTAL SALARY = SALARY + BENEFITS
5. Count the number of distinct DESIGNATIONs from EMPSALARY
6. List the employees whose names have exactly 6 characters
7. Add a new column PHONE_NO to the EMPLOYEE table and update the records
8. List employee names who have joined before 15-Jun-08 and after 16-Jun-07
9. Generate salary slip with Name, Salary, Benefits, HRA=50%, DA=30%, PF=12%,
and calculate gross salary. Order by gross salary in descending order.

Activity 3: (Logical, Relational Operators)

Database: Library

Create the following tables and insert tuples with suitable constraints.

Table: Books

Book_Id	Book_Name	Author_Name	Publishers	Price	Type	Quantity
C0001	The Klone and I	Lata Kappor	EPP	355	Novel	5
F0001	The Tears	William Hopkins	First Publ	650	Fiction	20
T0001	My First C++	Brain & Brooke	First Publ	350	Text	10
T0002	C++ Brainwork's	A.W. Rossaine	TDH	350	Text	15
F0002	Thunderbolts	Ana Roberts	First Publ	750	Fiction	50

Table: Issued

Book_Id	Quantity_Issued
T0001	4
C0001	5
F0001	2
T0002	5
F0002	8

Queries

1. To show Book name, Author name and price of books of First Publ. publisher.
2. Display Book id, Book name and publisher of books having quantity more than 8 and price less than 500.
3. Select Book id, book name, author name of books which is published by other than ERP publishers and price between 300 to 700.
4. Generate a Bill with Book_id, Book_name, Publisher, Price, Quantity, 4% of VAT "Total".
5. Display book details with book id's C0001, F0001, T0002, F0002 (Hint: use IN operator).
6. Display Book list other than, type Novel and Fiction.
7. Display book details with author name starts with letter "A".
8. Display book details with author name starts with letter "T" and ends with "S".
9. Select Book_Id, Book_Name, Author Name , Quantity Issued where Books.Books_Id = Issued.Book_Id.
10. List the book_name, Author_name, Price. In ascending order of Book_name and then on descending order of price.

Activity 4: (Date Functions)**Database: Lab**

Create Following table and insert tuples with suitable constraints.

Table: Equipment Details

No.	Item Name	Cost Per Item	Quantity	Date of Purchase	Warranty	Operational
1	Computer	30000	9	21/5/07	2	7
2	Printer	5000	3	21/5/06	4	2
3	Scanner	8000	1	29/8/08	3	1
4	Camera	7000	2	13/6/05	1	2
5	UPS	15000	5	21/5/08	1	4
6	Hub	8000	1	31/10/08	2	1
7	Plotter	25000	2	11/1/09	2	2

Queries

1. To select the ItemName purchase before 31/10/07.
2. Extend the warranty of each item by 6 months.
3. Display ItemName , Dateof purchase and number of months between purchase date and present date.
4. To list the ItemName in ascending order of the date of purchase where quantity is more than 3.
5. To count the number, average of costperitem of items purchased before 1/1/08.
6. To display the minimum warranty , maximum warranty period.
7. To Display the day of the date , month , year of purchase in characters.
8. To round of the warranty period to month and year format.
9. To display the next Sunday from the date "07-JUN-96".
10. To list the ItemName, which are within the warranty period till present date.

Part B:

Activity 5: (Numeric, character functions).

Use Functions for the following.

1. Find the mod of 165,16.
2. Find Square Root of 5000.
3. Truncate the value 128.3285 to 2 and -1 decimal places
4. Round the value 92.7683 to 2 and -1 decimal places.
5. Convert the string 'Department' to uppercase and lowercase.
6. Display your address convert the first character of each word to uppercase and rest are in lowercase.
7. Combine your first name and last name under the title Full name.
8. A) Take a string length maximum of 15 displays your name to the left. The remaining space should be filled with '*'.
B) Take a string length maximum of 20 displays your name to the right. The remaining space should be filled with '#'.
C) Take a string length maximum of 25 displays your name to the right. The remaining space should be filled with '@'.
9. Take a string length maximum of 20 displays your name to the right. The remaining space should be filled with '#'.
D) Take a string length maximum of 25 displays your name to the right. The remaining space should be filled with '@'.
10. Find the length of the string 'JSS College, Mysore'.
11. Display substring 'BASE' from 'DATABASE'.
12. Display the position of the first occurrence of character 'o' in Position and Length.
13. Replace string Database with Data type.
14. Display the ASCII value of ' ' (Space).
15. Display the Character equivalent of 42.

Activity 6: Database: subject

Create the following table and insert tuples with suitable constraints.

Table: Physics

Regno	Name	Year	Combination
AJ00325	Ashwin	First	PCM
AJ00225	Swaroop	Second	PMCs
AJ00385	Sarika	Third	PME
AJ00388	Hamsa	First	PMCs

Table: Computer Science

Regno	Name	Year	Combination
AJ00225	Swaroop	Second	PMCs
AJ00296	Tejas	Second	BCA
AJ00112	Geetha	First	BCA
AJ00388	Hamsa	First	PMCs

Queries

1. Select all students from Physics and Computer Science.
2. Select student common in Physics and Computer Science.
3. Display all student details who are studying in the second year.
4. Display students who are studying both Physics and Computer Science in the second year.
5. Display the students studying only Physics.
6. Display the students studying only Computer Science.
7. Select all students having PMCs combination.
8. Select all students having BCA combination.
9. Select all students studying in the third year.
10. Rename table Computer Science to CS.

Activity 7: (views)**Database:** Railway Reservation System.

Create the following table and insert tuples with suitable constraints.

Table: Train Details

Train_No	Train_Name	Start_Place	Destination
RJD16	Rajdhani Express	Bangalore	Mumbai
UDE04	Udhyan Express	Chennai	Hyderabad
KKE55	Karnataka Express	Bangalore	Chennai
CSE3	Shivaji Express	Coimbatore	Bangalore
JNS8	Janashatabdi	Bangalore	Salem

Table: Availability

Train_No	Class	Start_Place	Destination	No_of_seats
RJD16	Sleeper Class	Bangalore	Mumbai	15
UDE04	First Class	Chennai	Hyderabad	22
KKE55	First Class AC	Bangalore	Chennai	15
CSE3	Second Class	Coimbatore	Bangalore	8
JNS8	Sleeper Class	Bangalore	Salem	18

Queries

- Create a view **sleeper** to display train number, start place, destination which have sleeper class and perform the following:
 - Insert new record.
 - Update destination='Manglore' where train no='RJD16'.
 - Delete a record which has train no='KKE55'
- Create a view **detail** to display train number, train name, and class.
- Create a view **total_seats** to display train number, start place, use COUNT function on No_of_seats, group by start place and perform the following:
 - Insert new record.
 - Update start place='Hubli' where train no='JNS8'.
 - Delete the last row of the view.
- Rename view **sleeper** to **class**.
- Delete view **details**.

Activity 8: (group by, having clause)

Database: Bank system

Create the following table and insert tuples with suitable constraints.

Table: Account

Account_No	Cust_Name	Branch_ID
AE0012856	Reena	SB002
AE1185698	Akhil	SB001
AE1203996	Daniel	SB004
AE1225889	Roy	SB002
AE8532166	Sowparnika	SB003
AE8552266	Anil	SB003
AE1003996	Saathwik	SB004
AE1100996	Swarna	SB002

Table: Branch

Branch_ID	Branch_Name	Branch_City
SB001	Malleshwaram	Bangalore
SB002	MG Road	Bangalore
SB003	MG Road	Mysore
SB004	Jayanagar	Mysore

Table: Depositor

Account_No	Branch_ID	Balance
AE0012856	SB002	12000
AE1203996	SB004	58900
AE8532166	SB003	40000
AE1225889	SB002	150000

Table: Loan

Account_No	Branch_ID	Balance
AE1185698	SB001	102000
AE8552266	SB003	40000
AE1003996	SB004	15000
AE1100996	SB002	100000

Queries

1. Display the total number of accounts present in each branch.
2. Display the total loan amount in each branch.
3. Display the total deposited amount in each branch in descending order.
4. Display the maximum and minimum loan amount present in each city.
5. Display the average amount deposited in each branch for each city.
6. Display the maximum loan amount in each branch where the balance is more than 25000.
7. Display the total number of accounts present in each city.
8. Display all customer details in ascending order of branch ID.
9. Update the balance to 26000 where Account_No = AE1003996.
- 10.10. Display customer names with their branch name.

Evaluation Scheme for Lab Examination

Assessment Criteria Marks		Marks
Writing	One Program from Part A	15
	One Program from Part B	15
Execution	Any one of the Written Program	5
Viva Voce based on Data Base Management System		5
Total		40

Semester: III

Course Code: CAM33T	Course Title: Web Technologies
Course Credits: 03 (3-0-0)	Hours/Week: 03
Total Contact Hours: 44	Formative Assessment Marks: 20
Exam Marks: 80	Exam Duration: 03

Course Outcomes (COs):

CO1 Understand Internet basics, WWW, and HTML5 structure.

CO2 Design responsive web pages using HTML forms and CSS.

CO3 Write JavaScript code using variables, functions, and control structures.

CO4 Use JavaScript objects, arrays, events, and DOM manipulation.

Course Contents	Hours
Unit-1	
Foundations of Internet and World Wide Web: Introduction to the Internet, Internet Protocol Addresses, Domain Names, the World Wide Web, Web Browsers, Web Servers, Web Server Operation, General Server Characteristics, Uniform Resource Locators, Multipurpose Internet Mail Extensions, The Hypertext Transfer Protocol: -The Request Phase, The Response Phase, Security issues in WWW. Introduction to HTML5: Introduction, HTML Basics, Standard structure, HTML Elements, HTML Attributes, HTML Headings, HTML Paragraphs Text Formatting, Hyperlink and Images, Preserving White Space, Lists and Tables, HTML Block and Inline Elements, HTML Div Element, Horizontal Rules, Character Entities.	11
Unit-2	
HTML Form: Working with Form elements, Form attributes, Working with IFrames. Building Responsive Webpage. Cascading Style Sheets: Introduction, CSS Syntax, CSS Comments, Applying Styles to HTML Elements, Style specification formats, Selector forms, CSS Margins, CSS Padding, CSS Height, Width, CSS Text: Text Alignment, Text Decoration, Text Transformation, Text Spacing, Text Shadow, Font properties, List properties, The box model, CSS links, CSS List, CSS Position, CSS Z index	11

Unit-3	
Advanced CSS: CSS Rounded Corners, CSS Shadow Effects, CSS Text Effects, CSS 2D Transforms, CSS 3D Transforms, CSS Transitions, CSS Animations, CSS Masking. Introduction To JavaScript: Overview of JavaScript, General Syntactic Characteristics Advantages of JavaScript, variables, Comments, data types, constant, Screen Output & Keyboard Input, Operators, Type Conversion, Flow Controls: conditional statements, looping statements, jumping statements. Functions: Function Basic, function parameters, Function Invocation, Return statement, Global and local Variable.	11
Unit-4	
Object and Class: Definition of class, class syntax, class methods, constructor. Definition of object, object creation, object properties, built in objects: Arrays and strings: Definition of array, creation of array, Types of arrays, Accessing Array Elements, Array Properties and Methods, Strings: Definition, Creation of string, String Methods, Element Access in JavaScript. Events and Event Handling: Handling Events from Body Elements, Handling Events from Button Elements, Handling Events from Text Box and Password Elements, Element Visibility, Changing Colors and Fonts, Dynamic Content, Slow Movement of Elements, The navigator Object, Basics of Pattern Matching using RegExp.	11

Textbooks:

1. Robert W. Sebesta: Programming the World Wide Web, Pearson education.
2. Dr. T Vasudev, Dr. Chandrajit M, Arvind G and Vasanthi (2021), HTML 5, CSS 3 and JavaScript Made, Easy, Dream Books Publishing.
3. Thomas A Powell, Fritz schneider The Completer Reference: JavaScript Third Edition.

References book:

1. Thomas A. Powell: HTML & CSS: The Complete Reference, Fifth Edition.

Course Code: CAM33P	Course Title: Web Technologies Lab
Course Credits: 02 (0-0-2)	Hours/Week: 04
Total Contact Hours: 60	Formative Assessment Marks: 10
Exam Marks: 40	Exam Duration: 03

Course Outcomes (COs):

CO1 Create web pages using HTML tags, tables, lists, and forms.

CO2 Style web pages using inline, internal, and external CSS.

CO3 Use JavaScript for interactivity, validation, and event handling.

CO4 Build JavaScript programs for clocks, calculators, and animations.

Laboratory Program List

PART A

1. Develop and demonstrate HTML page containing basic text formatting tags, hyper link, images.
2. Develop and demonstrate HTML page containing III Sem BCA / III Sem B.Sc. [P.M. Cs] Course time table using table tag and its properties such as rowspan, colspan etc.
3. Develop HTML Page Contain College Course Details using Order and Unorder List
4. Design and develop a Student Application Form using HTML Form Elements
5. Design and develop HTML web page with appropriate tags to show the usage of
 - a. External level style specification.
 - b. Document level style specification.
 - c. Inline level CSS style specification.
6. Write an HTML program to create div and apply the following CSS properties on created div
 - a. Margin
 - b. Padding
 - c. Border Box shadow.
7. Write an HTML program to create a circle and create an animation of bouncing of the circle for 10 sec
8. Write HTML Program to Design Your College Name using follow CSS Properties
 - a. CSS Rounded Corners.
 - b. CSS Shadow Effects.
 - c. CSS Text Effects.

PART B

1. Develop a HTML-JavaScript program to find the factorial of an input number (Use Prompt window to accept input).
2. Design and develop HTML-JavaScript program to perform operations of a Simple calculator
3. Write a JavaScript program to implement any five string functions
4. Develop a HTML-JavaScript program to demonstrate the working of login page. (Display the messages in alert window).
5. Develop and demonstrate HTML page containing JavaScript function to change the font style of a paragraph on mouse over and on mouse out events.
6. Develop a HTML-JavaScript program to display Digital clock.
7. Develop a HTML-JavaScript program to validate University Register Number using Regular Expression.
8. Write a JavaScript program to hide and show the Image.

Evaluation Scheme for Lab Examination

Assessment Criteria Marks		Marks
Writing	One Program from Part A	15
	One Program from Part B	15
Execution	Any one of the Written Program	5
Viva Voce based on Web Technologies		5
Total		40

Semester: III

Course Code: CAE311(Elective)	Course Title: Cyber Security
Course Credits: 03(3-0-0)	Hours/Week: 03
Total Contact Hours:44	Formative Assessment Marks: 20
Exam Marks: 80	Exam Duration: 03

Course Outcomes (COs):

CO1 Describe cyberspace, internet infrastructure, and cyber security concepts.

CO2 Identify types of cybercrimes and understand related cyber laws.

CO3 Analyze social media risks, privacy issues, and legal aspects.

CO4 Use cyber security tools and apply best practices for protection.

Course Content	Hours
Unit 1	
Introduction to Cyber security: Defining Cyberspace and Overview of Computer and Web-technology, Architecture of cyberspace, Communication and web technology, Internet, World wide web, Advent of internet, Internet infrastructure for data transfer and governance, Internet society, Regulation of cyberspace, Concept of cyber security, Issues and challenges of cyber security.	11
Unit-2	
Cybercrime and Cyber law: Classification of cybercrimes, Common cybercrimes- cybercrime targeting computers and mobiles, cybercrime against women and children, financial frauds, social engineering attacks, malware and ransomware attacks, zero day and zero click attacks, Cybercriminals modus-operandi, Reporting of cybercrimes, Remedial and mitigation measures, Legal perspective of cybercrime, IT Act 2000 and its amendments, Cybercrime and offences, Organizations dealing with Cybercrime and Cyber security in India, Case studies.	11
Unit 3	
Social Media Overview and Security: Introduction to Social networks. Types of social media, social media platforms, social media monitoring, Hashtag, Viral content, social media marketing, social media privacy, Challenges, opportunities and pitfalls in online social network, Security issues related to social media, Flagging and reporting of inappropriate content, Laws regarding posting of inappropriate content, best practices for the use of social media, Case studies.	11
Unit 4	
Cyber Security Tools and Technologies: Introduction to antivirus, firewalls, intrusion detection and prevention systems (IDS/IPS), encryption tools, VPNs, and authentication mechanisms. Cyber Hygiene: Importance of strong passwords, regular updates, backups, and secure browsing habits. Network Security Fundamentals: Basics of securing wired and wireless networks, secure configurations, and network monitoring.	11

Text Books:

1. **Cyber Crime Impact in the New Millennium**, by R. C Mishra , Auther Press. Edition 2010.
2. **Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives** by Sunit Belapure and Nina Godbole,Wiley India Pvt. Ltd.

References Books:

1. **Compilers: Principles, Techniques, and Tools** by Alfred V. Aho, Monica S. Lam, Ravi Sethi, and Jeffrey D. Ullman, 2nd Edition, Pearson, 2007.
2. **Security in the Digital Age: Social Media Security Threats and Vulnerabilities** by Henry A. Oliver, Create Space Independent Publishing Platform. (Pearson , 13th November, 2001).
3. **Fundamentals of Network Security** by E. Maiwald, McGraw Hill.
4. **Cyber Laws: Intellectual Property & E-Commerce Security** by Kumar K, Dominant Publishers.

Semester: III

Course Code: CAE312 (Elective)	Course Title: Software Engineering
Course Credits: 03(3-0-0)	Hours/Week: 03
Total Contact Hours: 44 Hours	Formative Assessment Marks: 20
Exam Marks: 80	Exam Duration: 03

Course Outcomes (COs):

CO1 Explain software engineering concepts, process models, and agile practices.

CO2 Apply techniques to gather and manage software requirements.

CO3 Model systems using UML diagrams and system modeling techniques.

CO4 Design software architecture using patterns and UML notations.

Course Content	Hours
Unit 1	
Overview: Introduction; Software Engineering Ethics; Software Process Models; Process Activities; Coping with Change; Agile Software Development: Agile Methods; Plan- Driven and Agile Development.	11
Unit 2	
Requirements Engineering: Functional and Non-Functional Requirements; Software Requirements Document; Requirement's Specification; Requirements Engineering Processes; Requirement's Elicitation and Analysis; Requirement's Validation; Requirements Management.	11
Unit 3	
System Modeling: Context Models; Interaction Models- Use Case Modelling, Sequence Diagrams; Structural Models- Class Diagrams, Generalization, Aggregation, Behavioral Models- Data-Driven Modelling, Event-Driven Modelling; Model-Driven Engineering.	11
Unit 4	
Architectural Design: Architectural Design Decisions; Architectural Views; Architectural Patterns- Layered Architecture, Repository Architecture, Client-Server Architecture Pipe and Filter Architecture. Design And Implementation: Object-Oriented Design Using The UML- System Context and Interactions, Architectural Design, Object Class Identification, Design Models, Interface Specification; Design Patterns; Implementation Issues.	11

Text Books:

1. Software Engineering by Ian Sommerville, Pearson publications;10th Edition, 2015.
2. A Concise Introduction to Software Engineering, Pankaj Jalote, 2nd Edition, Springer Cham, 2025.

Reference:

1. Software Engineering, N.S. Gill, Khanna Publishing House, 2023.
2. Software Engineering A Practitioner's Approach, 8th edition, Roger S Pressman, Bruce R. Maxim. McGraw Hill Education, 2015.

Semester: III

Course Code: CAE313 (Elective)	Course Title: System Software
Course Credits: 03(3-0-0)	Hours/Week: 03
Total Contact Hours:44	Formative Assessment Marks: 20
Exam Marks: 80	Exam Duration: 03

Course Outcomes (COs):

CO1 Understand system software, SIC/SIC-XE, CISC and RISC.

CO2 Explain assembler functions and features.

CO3 Describe loader and linker operations.

CO4 Identify functions of DBMS, editors, and debuggers.

Course Content	Hours
Unit 1	
System software: Introduction, System Software and Machine Architecture, The Simplified Instructional Computer (SIC)- SIC Machine Architecture, SIC / XE Machine Architecture, SIC programming examples, Traditional (CISC) Machines and RISC Machines- Introduction, Differences.	11
Unit 2	
Assemblers: Basic Assembler Functions- A Simple SIC Assembler, Assembler Algorithm and Data Structures, Machine Dependent Assembler Features- Instruction Formats and Addressing Modes, Program Relocation, Machine Independent Assembler Features- Literals, Symbol Defining Statements, Expressions, Program Blocks, Control Sections and Program Linking.	11
Unit 3	
Loaders and Linkers: Basic Loader Functions- Design of an Absolute Loader, Machine Dependent Loader Features, Relocation, Program Linking, Machine Independent Loader Features- Automatic Library Search, Loader Options, Loader Design Options- Linkage Editors, Dynamic Linking, Bootstrap Loaders	11
Unit 4	
Other System Software: Database Management Systems- Basic Concepts of a DBMS, Levels of Data Description, Use of a DBMS, Text Editors- Overview of the Editing Process, User Interface, Editor Structure, Interactive Debugging Systems- Debugging Functions and Capabilities, Relationship with Other Parts of the System, User Interface Criteria.	11

Text Books:

- 1. System Software-** An Introduction to Systems Programming, 3rd Edition, Leland L. Beck, D Manjula, Pearson Education.
- 2. Systems Programming,** John J Donovan, McGraw Hill Education.

Reference Books:

- 1. Compilers: Principles, Techniques, and Tools** by Alfred V. Aho, Monica S. Lam, Ravi Sethi, and Jeffrey D. Ullman, 2nd Edition, Pearson, 2007.
- 2. Lex & Yacc** by Doug Brown, John Levine, and Tony Mason, O'Reilly Media, October 2012.

Semester: III

Course Code: CAE321 (Elective)	Course Title: Cloud Computing
Course Credits: 03(3-0-0)	Hours/Week: 03
Total Contact Hours:44	Formative Assessment Marks: 20
Exam Marks: 80	Exam Duration: 03

Course Outcomes (COs):

CO1 Understand cloud computing concepts, architecture, benefits, and challenges.

CO2 Explain cloud service models (IaaS, PaaS, SaaS) and their providers.

CO3 Describe deployment models, virtualization types, and hypervisors.

CO4 Explore cloud storage, databases, networking, and DevOps practices.

Course Content	Hours
Unit 1	
Introduction to Cloud Computing: Definition and Characteristics of Cloud Computing, History and Evolution of Cloud Computing, Cloud Computing Architecture, Benefits and Challenges of Cloud Computing.	11
Unit 2	
Cloud Service Models: Infrastructure as a Service (IaaS), Key Providers: AWS EC2, Google Compute Engine, Azure VMs, Platform as a Service (PaaS), Key Providers: AWS Elastic Beanstalk, Google App Engine, Azure App Services Software as a Service (SaaS), Examples: Google Workspace, Microsoft Office 365.	11
Unit 3	
Cloud Deployment Models: Public Cloud, Private Cloud, Hybrid Cloud, Community Cloud, Concept of Virtualization, Types of Virtualizations: Server, Network, Storage, Hypervisors: VMware, Hyper-V, KVM.	11
Unit 4	
Cloud Storage and Databases: Storage Solutions, S3, Azure Blob Storage, Google Cloud Storage, Database Services AS AWS RDS, Azure SQL Database, Google Cloud SQL, NoSQL Databases as DynamoDB, Azure Cosmos DB, Google Cloud Fire store. Cloud Networking: Networking Basics in the Cloud, Virtual Private Cloud (VPC), Load Balancing and Auto Scaling, Content Delivery Networks (CDN). Introduction to DevOps in the Cloud: DevOps Principles and Practices, CI/CD Pipelines, Infrastructure as Code (IaC).	11

Text Books:

1. "Cloud Computing: Concepts, Technology & Architecture" by Thomas Erl, PHI.

Reference Books:

1. AWS Documentation, Azure Documentation, Google Cloud Documentation, **Tools:** AWS Free Tier, Azure Free Account, Google Cloud Free Tier.

Semester: III

Course Code: CAE322 (Elective)	Course Title: Digital Image Processing
Course Credits: 03(3-0-0)	Hours/Week: 03
Total Contact Hours: 44	Formative Assessment Marks: 20
Exam Marks: 80	Exam Duration: 03

Course Outcomes (COs):

CO1 Understand the fundamentals of digital and color image processing.

CO2 Apply spatial and frequency domain techniques for image enhancement.

CO3 Implement image restoration using filtering and degradation models.

CO4 Perform image segmentation, morphological operations, and compression techniques.

Course Content	Hours
Unit 1	
Digital Image Processing: Definition, History, Applications Of DIP, Fundamental Steps In Digital Image Processing, Components Of An Image Processing System, Elements Of Visual Perception, Image Sensing And Acquisition, Simple Image Formation, Image Sampling And Quantization, Representing Digital Pixels, Image Quality. Introduction to Colour Image: Colour Fundamentals, RGB And HSI Models, Pseudo Colour Image Processing.	11
Unit 2	
Image Enhancement In Spatial Domain: Introduction to Image Enhancement, Basic Grey Level Transforms, Histogram, Histogram-Processing Equalization, Matching & Colour Histogram, Enhancement Using Arithmetic/Logic Operations, Spatial Filtering, Smoothing Spatial Filtering, Sharpening Spatial Filtering. Image Transform: Fourier Transform, SHFT, DFT, FFT, DCT, Hadamard Transform, Wavelets transform (CWT, DWT), KLT, SVD, Applications.	11
Unit 3	
Image Enhancement in frequency domain: Smoothing Frequency Domain Filtering, Sharpening Frequency Domain Filtering. Image Restoration: Model For Image Degradation/Restoration Process, Noise Model, Mean Filtering And Filtering, Estimating Degradation Function, Inverse Filtering, Minimum Mean Square Error (Wiener Filter), Colour Image Smoothing, Sharpening.	11

Unit 4	
Segmentation & Morphological operations: Segmentation And Threshold Function, Different Algorithms In Thresholding, Line Detection, Edge Detection, Edge Linking By Graph Search Method, Hough Transform, Region-Based Segmentation, Matching, Colour Segmentation, Morphological-Dilation And Erosion, Opening And Closing, Hit/Miss Transforms, Representation Boundary Descriptors, Regional Descriptors. Image Compression: Need For Image Compression, Huffman, Run-Length Encoding, Shift Codes, Vector Quantization, Transform Coding, JPEG Standard, MPEG.	11

Text Book:

1. Digital Image Processing- Rafael C Gonzalez and Richard E. Woods, PHI 3rd Edition 2010.
2. K. Jain "Fundamental of digital image processing", Prentice-Hall, 2002.

Reference:

1. R. C. Gonzalez, R. E. Wood "Digital image processing using MATLAB", Pearson Education, 2004.
2. M. Sonka, V. Hlavac, R. Boyle, "Image processing analysis and machine vision" Chapman & Hall, 1998.
3. Digital Image Processing- S. Jayaraman, S. Esakkirajan, T. Veerakumar, Tata McGraw Hill 2014.

Semester: III

Course Code: CAE323 (Elective)	Course Title: E-commerce and E-Governance
Course Credits: 03(3-0-0)	Hours/Week: 03
Total Contact Hours: 44	Formative Assessment Marks: 20
Exam Marks: 80	Exam Duration: 03

Course Outcomes (COs):

CO1 Understand the fundamentals, models, and infrastructure of E-commerce.

CO2 Analyze E-commerce strategies, marketing, and real-world applications.

CO3 Explain E-Governance models, technologies, and service frameworks.

CO4 Evaluate challenges, innovations, and future trends in E-Governance.

Course Contents	Hours
Unit-1	
Introduction to E-commerce: Fundamentals of E-commerce, E-commerce Business Models, E-commerce Infrastructure, E-commerce Security and Payment Systems	11
Unit-2	
E-commerce Strategies and Applications: E-commerce Marketing and Advertising, Supply Chain Management in E-commerce, Emerging Trends in E-commerce, Case Studies and Practical Applications.	11
Unit-3	
Introduction to E-Governance: Fundamentals of E-Governance, E-Governance Models and Frameworks, Technology and Infrastructure for E-Governance, E-Governance Services and Applications.	11
Unit-4	
Challenges and Future Trends in E-Governance: Challenges in E-Governance, E-Governance in Developing Countries, Emerging Technologies in E-Governance, Future Directions and Innovations.	11

Textbooks:

1. "E-Commerce 2020: Business, Technology, and Society" by Kenneth C. Laudon and Carol Guercio Traver, Pearson.
2. "Electronic Commerce 2018: A Managerial and Social Networks Perspective" by Efraim Turban, Jon Outland, David King, Jae Lee, Ting-Peng Liang, and Deborrah C. Turban, Springer.

Reference Books:

1. "E-Government: Information, Technology, and Transformation" by Hans J. Scholl, Routledge.
2. "E-Governance: Managing or Governing?" by Jeremy Millard, Routledge.
3. "Public Information Technology and E-Governance: Managing the Virtual State" by G. David Garson, Jones & Bartlett Learning.